1: Ethical Business Plan

1.A. Company Name

CerebraTech

1.B. Long-Term Vision Statement

1.B.1 Goals

The primary goal at CerebraTech is to become an industry leader in non-invasive neural prosthetic technology by delivering highly accurate, responsive, and accessible systems across socioeconomic and demographic groups. However, our main focus is on adult amputees between the ages of 18 and 70, including veterans, individuals with diabetes-related amputations, and trauma survivors. CerebraTech also aims to expand into broader technology applications such as rehabilitation, assistive communication, and enhanced human-computer interaction while maintaining safety, equity, and ethical innovation.

1.B.2 Idea Origination

The idea for CerebraTech came from our desire to provide a solution to real-world healthcare problems that we can use our computer science skills for. We were also interested in the advancement of prosthetic technology that utilizes neural signals to coordinate movement in prosthetic limbs for amputee patients. At the same time, we wanted to provide effective and accessible solutions for underserved populations, such as veterans and low-income patients.

1.B.3 Purpose/Values/Mission

CerebraTech's purpose is to empower individuals with physical disabilities by restoring autonomy and improving their quality of life through computer-brain interface technology. The mission of the company is to design safe, reliable, and equitable neural interface systems that prioritize user privacy, accessibility, and dignity. CerebraTech's core values include responsible innovation, inclusivity, and collaboration with our primary stakeholders.

1.B.4 Key Questions

CerebraTech will guide its development using the following key questions:

- How can CerebraTech maximize positive impact on patients' independence and quality of life while minimizing risk?

- How can CerebraTech ensure its technology remains accessible and equitable across different income levels, races, and backgrounds?
- How can CerebraTech innovate rapidly in the field of brain-computer interfaces while maintaining the highest standards of data privacy, security, and regulatory compliance?

1.C. Strategy with Ethical Impacts and Ethical Safeguards

1.C.1.1 OKR 1 Objective and Key Result

CerebraTech will create safe and reliable rules for handling all neural data collected by its brain-computer interface systems within 12 months of their first use. This can be done by ensuring that all neural data is fully protected with encryption, using the AES-256 encryption method for stored data and the TLS 1.3 protocol for data being sent over CerebraTech's network. We will also conduct at least two independent audits from third-party cybersecurity firms with no critical vulnerabilities found and maintain at least an 85% user trust score regarding our data security practices. Our primary customer base's demographics may include a wide range of people from all ethnic and racial backgrounds, income levels, and ages. Key stakeholders would include the patient, the patient's doctors, insurance providers, CerebraTech engineers, and third-party cybersecurity firms. Patients will rely on the secure handling of highly sensitive neural and personal information, while their doctors need access to patient performance data while maintaining confidentiality. Insurance providers will need to evaluate the cost-effectiveness of our product to provide coverage for patients, and CerebraTech engineers will be responsible for designing and handling data generated by the system. Lastly, cybersecurity firms would be responsible for conducting independent audits to verify our system integrity.

1.C.1.2 OKR 1 Metrics with Experimentation

We can measure the success of our OKR across a few metrics. By doing internal system audits on all of our data pipelines and finding all of our storage endpoints and possible transmission channels, we can see what percentage of our total data is protected by the required encryption standards. The goal is 100%. Another metric is the number and severity of vulnerabilities found in third-party audits, which can be obtained by hiring two external cybersecurity firms for penetration testing, aiming for no critical vulnerabilities. The last metric we could use to determine the success of our OKR is to conduct a user trust survey among CerebraTech customers on a scale of 0-100 with questions such as

1. I trust CerebraTech to securely handle my neural data.
2. I understand how my data is used and stored.
3. I feel confident my data will not be misused.

4. I am comfortable sharing my data for system improvements.

We could then average the percentage across responses to these questions, with the target being at least 85%.

1.C.1.3 OKR 1 Ethical Impacts/Issues

Some potential ethical impacts that could arise from this OKR are the privacy risks associated with the neural signals collected as part of the computer-brain interface, as a data breach could expose deeply personal information related to cognitive or emotional function. Users may not fully understand the collection, storage, or usage of their personal or neural information. These issues could stop them from giving their full and informed consent for data use. These problems relate to the case of Carpenter v. United States, as the government obtained information from cellphone companies without any warrants establishing probable cause to collect such information [1], meaning that the privacy of patients involved with the technology could be violated by law enforcement if they search any information associated with CerebraTech technology.

Expected Ethical Impact Risk Table

Stakeholder	Financial Risk	Privacy Risk	Conflicting Interest Risk	Violation of Rights Risk
Patient	Mid	High	Mid	High
Doctors	Mid	Mid	Low	Low
Insurance Providers	Mid	Low	High	Low
CerebraTech	High	High	Mid	Mid

- Patient: The financial risk for patients is moderate because they may need to pay out-of-pocket if insurers deny coverage for the prosthetic system due to potential security failures. The privacy risk is high because the system would collect very sensitive neural data that could show personal cognitive states, emotional patterns, or intentions. The conflicting interest risk is moderate because patients will want a product that has the maximum amount of privacy and control, while CerebraTech may want to collect data for improving the system. The risk of violating rights is high because consent could be unclear and the data could be misused, which may violate the patient's rights to privacy and bodily autonomy.

- Doctors: Doctors' financial risks are moderate because they could face some liability if they prescribe systems that are insecure and harm patients. The privacy risks are moderate because doctors must comply with HIPAA and keep patient data confidential. The conflicting interest risk is low because doctors may prioritize patient well-being over profit motives. The risk of violating rights is low because any misconduct by CerebraTech will not necessarily affect the doctors.
- Insurance Providers: The financial risk for insurance providers is moderate because they could lose money if they cover brain-computer interface systems that break or need to be replaced. The privacy risk for insurance providers is low because they only access summaries of data rather than raw data, so the risk of exposure is lower. The risk of conflicting interests is high because insurance providers may prioritize cost reduction over patient access to prosthetics. The risk of violating rights is low because choosing whether or not to provide coverage to patients is within their rights.
- CerebraTech: CerebraTech has a high financial risk because any data breaches could result in lawsuits and reputational damage. The privacy risk is also high because CerebraTech is responsible for all data that passes through the computer-brain interface system, and any failures are the responsibility of the company. The risk of a conflicting interest is moderate because CerebraTech has to collect data to improve future models of the computer-brain interface system while also following privacy limitations. The violation of rights risk is moderate because improper data use or lack of patient consent can violate the patient's rights.

1.C.1.4 OKR 1 Ethical Safeguards

The first safeguard we will use is end-to-end encryption and a zero-trust security architecture to mitigate the risks associated with data privacy. Cybersecurity experts and compliance officers, whom we could hire from external firms, will design this safeguard. We would implement this safeguard with continuous monitoring systems in our network, such as intrusion prevention systems or intrusion detection systems, and quarterly penetration testing of the company networks. We can measure the effectiveness of this safeguard with the reports gathered from the quarterly penetration testing and auditing of the encrypted data. This is in line with Section 1.7 of the Association for Computing Machinery's Code of Ethics [2], as this supports the defense of confidentiality for our end users. The second safeguard we would use to ensure that patients are aware of the data collected and used by CerebraTech systems is maintaining a transparent data governance framework that provides users with dashboards on their devices that show what data is being collected and how CerebraTech intends to use it, while also allowing users to delete or restrict data as they wish. We can design this safeguard with the help of ethical experts to determine what data is essential for us to keep and what can be restricted by the user. We can measure the efficacy of this safeguard by surveying how much users trust our data transparency and other metrics on how much the end user checks the data collected by CerebraTech.

1.C.2.1 OKR 2 Objective and Key Result

Objective: The Cerebratech device will process each user's unique brain signals with a machine learning model that continuously adapts to that individual over time. Patients are expected to show a measurable improvement in prosthetic control over the first six months of adapting.

Key Result: At least 90% of active Cerebratech users will show a statistically significant improvement in functionality between the beginning and end of the adapting and learning period. Functionality will be measured by the number of tasks/functions the user can complete and with what level of difficulty the task is completed or the level of assistance needed to complete the task.

The core idea is that no two people's brains produce identical neural patterns for the same intended movement, and those patterns can also shift over time due to neuroplasticity, fatigue, or mental state. A generic, fixed model would therefore produce inconsistent or frustrating results for many users. The machine learning model continuously learns movement signals paired with actual prosthetic outcomes to tune the decision boundary for each individual profile.

The stakeholders most directly affected by this OKR are patients themselves. The patient population for Cerebratech spans a wide demographic range. Limb loss affects adults of all ages, especially military veterans, and adults with vascular disease or diabetes. Limb loss is agnostic to race and is present among all classes. This demographic spread is significant because it is unclear to what extent these variables indirectly affect the relevant neural patterns. The machine learning model must be trained on data that reflects this diversity; otherwise, its personalization may work well for some groups and poorly for others.

Families and caregivers are indirect stakeholders. If the device works reliably, it reduces their caregiving burden. Case clinicians are participants in the OKR because they give dexterity assessments and interpret progress. Payers are financial stakeholders who will want to see the six-month improvement as a deciding factor for supporting coverage of the program. Regulatory bodies are legal stakeholders that will look for evidence of effectiveness and safety before granting approval. Cerebratech has a commercial interest in this OKR, but also a fiduciary interest in ensuring the model's improvements are genuine and not artificially skewed or biased.

1.C.2.2 OKR 2 Metrics with Experimentation

To measure utility and user functionality, two metrics will be used: Task Independence Level and Task Difficulty Rating. These two metrics measure if a user can complete a task and how hard the task is. A 5 point ranking system can quantify the independence level with a range of 1 “completely dependent” to 5 “requires no assistance”. A similar ranking system can be used for difficulty with 1 being “extremely frustrating/near impossible”, to 5 being “completely natural, requires little concentration”. This could be further expanded to capture emotional, mental, and physical response by asking the level of concentration required or how fatiguing the task was. Timing the length of the task may also be a secondary indicator of task difficulty.

A standardized test suite of everyday tasks would serve as a litmus test to measure patient functionality. For example, using a utensil to eat, getting into and out of bed, putting on clothes, opening a door, holding a shopping bag, etc. The selection of tasks to be evaluated would be individually curated based on the disability.

1.C.2.3 OKR 2 Ethical Impacts and Issues

A primary ethical risk is an algorithmic bias in the machine learning model. If the training data used to seed the model is not representative of the full diversity of the user population, the adaptation process may converge more slowly or less completely for underrepresented groups. In practice, this could mean that a White male military veteran whose demographic is overrepresented in early beta testing achieves excellent performance within weeks, while an older African American woman with a diabetic amputation lags behind due to differing neural signals and causation for limb-loss.

A real-world parallel to this risk is found in the case of pulse oximetry bias. Research published in the New England Journal of Medicine found that FDA-cleared pulse oximeters, devices that estimate blood oxygen saturation, overestimated oxygen levels in Black patients compared to White patients at a significant rate. This flaw was rooted in the devices being calibrated primarily on lighter-skinned subjects [3]. Due to the countless confounding variables that may affect neural signaling, it is critical that data is collected from all types of race, ethnicities, lifestyles, and medical histories.

A second ethical issue is data privacy. The personalization process requires the device to continuously collect the user's neural signals, map them to intended movements, and store this data to update the model. Neural data is sensitive. It can potentially reveal motor intentions, emotional states, cognitive patterns, or medical conditions beyond the scope of limb control if collected in volume. Even if raw neural data never leaves the device, the behavioral patterns inferred from it could be commercially valuable and could be targeted by malicious actors. If a user opts into sharing data with Cerebratech for broader model improvement, anonymization provides partial protection to the user. However, even if the method of how to use the anonymized data for financial gain is not evident, “Big Data” is always in high demand and can be sold for a price. For this purpose, user data will never be sold to third parties.

A third issue is consent. The continuous, passive nature of neural data collection means users may not fully understand the extent of the data being gathered. Informed consent for the device is more complex than for a standard medical device because the data being collected comes directly from the brain. Vulnerable populations, such as older adults with cognitive decline or minors, may face challenges in providing meaningful informed consent.

Expected Ethical Impact Risk Table

Stakeholder	Financial Risk	Privacy Risk	Conflicting Interest Risk	Violation of Rights Risk
Patient/User	Mid	High	Mid	High
Cerebratech	High	Mid	High	Low
Clinicians	Low	Low	Mid	Low
Payers/Insurers	High	Low	High	Low
Regulators (FDA)	Low	Low	Mid	Mid

OKR 2 Analysis of Ethical Impact Risk

Patient/User Stakeholder: Financial risk is rated mid because users who are less receptive to the device need additional clinical visits and receive diminished utility from the product. Privacy risk is high because the device collects neural signals on a continuous basis. A breach or unauthorized use of this data could expose personal information about the user. Conflicting interest risk is medium because patients want a device that maximizes their personal functional improvement, while Cerebratech's model improvements may prioritize majority-group accuracy. Violation of rights is high- If underrepresented groups receive inferior adaptation, they are being denied equal benefit from the technology, which constitutes a violation of equitable access rights.

Cerebratech Stakeholder: Financial risk is high because of product performance, if it falls short could trigger regulatory action, tarnished reputation, product recalls, or lawsuits- any of which could be catastrophic for a startup. Privacy risk is mid because Cerebratech handles anonymized neural data on its servers, but needs to protect it from attacks. Conflicting interest risk is high because the company's incentive is to ship a product quickly and demonstrate impressive performance metrics which can conflict with the ethical priorities. Violation of rights risk is low at the company level because the company itself does not have rights that are being violated.

Clinicians Stakeholder: Financial risk is low because clinicians are paid for their services regardless of device performance. Privacy risk is low because clinicians have extremely limited access to relevant data. Conflicting interest risk is mid because clinicians want the best outcome for each patient but may feel financial pressure or incentive from payers or Cerebratech to endorse the device. Violation of rights risk is low.

Payers/Insurers Stakeholder: Financial risk is high because payers who cover device costs are directly exposed to all costs associated with the device. Conflicting interest risk is high because payers want to minimize expenditure while patients and clinicians want the best possible care regardless of cost. Privacy risk and violation of rights risk are both low for payers since they are not exposed to it.

Regulators (FDA) Stakeholder: Financial risk is low because Cerebratech has no financial impact on the regulators. Privacy risk is low since the FDA is a trusted government organization. Conflicting interest risk is low because the FDA has no financial incentive to act a certain way. Violation of rights risk is mid because if the FDA approves a device with demographic bias, it fails to mandate inclusive testing.

1.C.2.4 OKR 2 Ethical Safeguards

The most important safeguard against algorithmic bias is what we can call a Demographically Stratified Data Collection Protocol. Before the device is ever submitted to the FDA for approval, the machine learning model must be trained on data that actually represents the full range of people who will use it (as best as possible). To achieve this, Cerebratech will recruit a training dataset of participants with diverse representation, age, sex, race, ethnicity, lifestyle, and cause of amputation to contribute a foundation of labeled data. This directly follows guidance from the FDA's 2022 Action Plan for Racial Equity in Medical Devices, which calls on device manufacturers to proactively test and verify that their products perform equally well across racial and ethnic groups [4].

The rollout would happen in phases. First, Cerebratech would collect and label training data across partner clinical sites. The model would then be trained on that data and model training success will be determined by equal functionality across diversity metrics. Data will continue to be collected to close the gap between lagging groups. Finally, the model will enter real world trials where the fairness checks will continue to be run on live user data.

To protect user privacy during the ongoing personalization process, Cerebratech machine learning models are periodically trained on user data, but model state/checkpoint is deleted after updating to the user device. The only persistent copies of the models are always only on the user device. This means there is no central location where anyone could piece together an individual user's neural data.

Cerebratech will address the issue of informed consent through the Cerebratech mobile app. Instead of a long legal document, the app will walk through short, plain-language explanations of what data is collected, where it goes, who can see it, and how to opt out. This encourages consent that is genuine.

1.C.3.1 OKR 3 Objective and Key Result

CerebraTech's third OKR will focus on equitable access and demographic inclusion for all patients. Our objective will be to ensure that the brain-computer interface system performs equally well across the full range of people who will use it and not just most. The key results we are looking for: at least 40% of beginning participants will come from underrepresented racial and ethnic groups, model performance variance across demographic groups will be kept at 5% or lower, and CerebraTech will establish partnerships with at least three public hospitals that will help low-income areas. The main idea behind this is that a device that only works well for one type of person is not what we are looking for, regardless of what the overall performance numbers are, because people who have disabilities are across all racial, ethnic, and income backgrounds, so making sure we can fit most people is important to us.

The main stakeholders will be patients, whom we would look out for in underrepresented communities. Other stakeholders would be clinical partners like the FDA and insurance providers, who will use performance data to make their decisions on coverage.

1.C.3.2 OKR 3 Metrics with Experiments

For our first metric, we want to see participant demographics and will track the racial and ethnic breakdown of all participants against disabled and amputee population data, targeting at least 40% from underrepresented groups. We'll find these patients through our partnership with public hospitals and other disabled and amputee organizations that already serve diverse patient populations. Next, we are looking to model performance variance across demographic groups. After each training and testing cycle, we will run performance reports that are broken down by race, ethnicity, age, sex, and cause of amputation, with the target being that there isn't a group's average performance that falls more than 5% below the overall average. If there is a group that is larger than the 5%, we then have a mandatory review of the training data and model architecture before moving forward. Lastly, we want to track the number of active partnerships with public hospitals, targeting at least three in the first year of the program.

1.C.3.3 OKR 3 Ethical Impacts/Issues

The main ethical risk, as discussed in OKR2 is again, algorithmic bias, meaning that if the training data used to develop the model is not representative of the full user population, then the device will systematically underperform for certain groups. As previously mentioned, Research published in the New England Journal of Medicine found that FDA-cleared pulse oximeters consistently overestimated the blood oxygen levels in Black patients compared to white patients. A similar failure in a neural prosthetic system would mean that underrepresented patients will receive the prosthetic that isn't fully functional for some, but others work perfectly, which would be an ethical failure on our part and could also be a violation of equitable access [6].

Expected Ethical Impact Risk Table

Stakeholder	Financial Risk	Privacy Risk	Conflicting Interest Risk	Violation of Rights Risk
Patient/User	Mid	High	Mid	High
CerebraTech	High	Mid	High	Low
Clinicians	Low	Low	Mid	Low
Insurance	High	Low	High	Low
FDA	Low	Low	Mid	Mid

1.C.3.4 OKR 3 Ethical Safeguards

The most important safeguard we are looking at is that before our submission to the FDA, we will recruit a training dataset that will reflect the full diversity of the expected user population across age, sex, race, ethnicity, and cause of needing our device. We'll follow the FDA's 2022 Action Plan for Racial Equity in Medical Devices, which requires manufacturers verify that their products perform equally well across racial and ethnic groups [5]. We will also put our testing program in hospitals that already serve low-income and underinsured patients, ensuring that the people most at risk of being excluded are being included in the development process, even without being able to afford the device. We are also going to gather performance data that will continue to be collected and reviewed on a quarterly basis, and if performance gaps show, we will continue to update the model to close the gap and improve the functionality.

2: Cultural Policy

2.A. Core Values

CerebraTech seeks to be known as a company that puts human dignity at the center of technological innovation. We want to be recognized for developing technology that genuinely improves our clients' lives while respecting their rights and identities. Our core values center on ethical responsibility so that every design decision is made with careful consideration of privacy, consent, and long-term societal impact. We value inclusivity, ensuring that our systems are designed to work effectively for people from diverse backgrounds, regardless of their race, gender, age, or income level, so that access to our technology is not limited to only those with financial privilege.

We also prioritize transparency in how we collect, process, and store sensitive neural data so that users can always understand how their information is being used and have control over it. Collaboration is another of our core values, since we believe that the best solutions for problems that people face come from partnerships between engineers, doctors, patients, and regulators. We also emphasize accountability by holding ourselves responsible for any failures, learning from these failures, and continuously improving.

2.B. Motivation

CerebraTech is driven by our love for the idea that technology can help restore a person's dignity and agency by allowing them to perform everyday tasks, regain confidence, and reconnect with the world around them. We are motivated by the opportunity to work on innovations that have a clear impact on people and help resolve problems with real-world healthcare solutions. However, we are also guided by the fear of the potential for harm that may come to our users through system failures, security breaches, or misuse of their neural data. We also fear contributing to inequality by allowing only certain groups to benefit from our innovative technologies and also losing public trust, as even a single failure could be detrimental to the confidence in our company and the field of prosthetics.

2.C. Summary

1. Ethical
2. Innovation
3. Empowering
4. Human Potential
5. Equity

3: Ethics Policy

3.A. Core Items

To ensure that CerebraTech fulfills its mission while maintaining the safety standards and ethics, the following policies have been outlined:

- **Data Privacy**: This policy requires that all data collected by CerebraTech devices are the exclusive property of the user. To mitigate the risk of a data breach, our architecture is designed so that persistent machine learning models exist only on the user's local device. Central model states are deleted after individual updates.
- **Demographic Accuracy**: The model performance across different races, ethnicities, ages, and sexes must remain comparable. This prevents algorithmic bias seen in previous medical technologies, like pulse oximetry, where lighter skin tones were prioritized in calibration.
- **Informed Consent**: Our policy requires the use of a simple, mobile interface that explains exactly what data is being collected and allows users to delete or restrict data at any time.
- **Equitable Access**: To keep the product accessible, CerebraTech will maintain partnerships with public hospitals serving low-income and underinsured communities. This is to promote the inclusion of people most at risk of being excluded from technological advancement.
- **Commercialization of Data**: CerebraTech will never sell anonymized or raw neural data to third parties for commercial gain. Our responsibility is to the patient’s privacy, not to data markets.

3.B. Board

The first board member chosen is Dr. Timnit Gebru, a widely recognized computer scientist and the founder of the Distributed AI Research Institute (DAIR). Dr. Gebru is a leading expert in algorithmic bias and data ethics and the risks of large scale machine learning models. Her position on the board is ideal because CerebraTech’s functionality relies on adaptive machine learning models that must interpret neural signals across diverse populations. Dr. Gebru’s expertise will help prevent the exclusion of minority groups, such as the discussed case of the pulse oximeters.

The second board member is Dr. Alondra Nelson, Professor at the Institute for Advanced Study and former director of the White House Office of Science and Technology Policy. Dr. Nelson was an architect of “Blueprint for an AI Bill of Rights”, which protects against malicious data practices and discrimination in healthcare and tech.Dr. Nelson’s career is perfectly centered at the intersection of science, technology, and social equality. Her experience in ethics will help ensure that our systems are equitable and that patients will have control over their biological data. This all will ultimately push towards the goal to provide accessible solutions for underserved populations and low-income patients.

The third board member is Palmer Luckey, the founder of Oculus VR and Anduril Industries. His foundational experience in developing the first commercially viable head mounted displays and motion tracking hardware gives him invaluable insight in human to computer interaction. Luckey understands the technical challenges of low-latency signal processing and the hardware-software integration required for bionic prosthetics. His experience in scaling a tech startup from a core idea to an industry leader will provide CerebraTech the guidance needed to transition from clinical trials to a widely accessible consumer product while keeping high-performance, responsive systems.

4: YouTube Presentation

https://youtu.be/nEh71kmtiBM

5: References

[1] ACLU. 2018. *Carpenter v. United States*. Retrieved from https://www.aclu.org/cases/carpenter-v-united-states.

[2] ACM. 2018. ACM Code of Ethics and Professional Conduct. Retrieved from https://www.acm.org/code-of-ethics.

[3] Sjoding, M.W., Dickson, R.P., Iwashyna, T.J., Gay, S.E., and Valley, T.S. 2020. Racial bias in pulse oximetry measurement. New England Journal of Medicine. 383, 25, 2477–2478. DOI: https://doi.org/10.1056/NEJMc2029240

[4] U.S. Food and Drug Administration. 2022. FDA Action Plan for Racial Equity: Medical Devices. U.S. Department of Health and Human Services. Retrieved from https://www.fda.gov/regulatory-information/search-fda-guidance-documents/diversity-action-plans-improve-enrollment-participants-underrepresented-populations-clinical-studies

[5] Zuckerman, D., Jury, N., and Silcox, C. 2018. Diversity in medical device clinical trials: do we know what works for which patients? *Milbank Quarterly*. 96, 3, 499–529. https://pmc.ncbi.nlm.nih.gov/articles/PMC6131322/

[6] Burwell, S., Sample, M., and Racine, E. 2017. Ethical aspects of brain computer interfaces: a scoping review. *BMC Medical Ethics*. 18, 60. https://pmc.ncbi.nlm.nih.gov/articles/PMC5680604/